Did Massachusetts' 1992 Tobacco Control Initiative Reduce Cigarette Sales?

Evidence from Synthetic Control and Synthetic Difference-in-Differences

Qiaoshi Yang
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Abstract

In November 1992 Massachusetts voters approved Question 1, which raised the cigarette excise tax by 25 cents per pack and used the revenue to fund a statewide tobacco control program. I estimate its effect on per-capita cigarette sales with the synthetic control method of Abadie et al. (2010), after first replicating their estimate for California's Proposition 99 with the same code. Synthetic Massachusetts, a weighted average of Connecticut, Delaware, Utah and Rhode Island, tracks Massachusetts closely from 1970 to 1992. After 1992 the two diverge: by 2000 actual sales are 29 packs per capita below the synthetic counterfactual, and the average gap over 1993–2000 is 18.5 packs, or 21% of counterfactual sales. Massachusetts has the third-largest post/pre fit ratio among 39 states (permutation $p = 0.077$) and the largest standardised reduction ($p = 0.026$). Two-way fixed effects DiD and synthetic DiD give average effects of -19 and -17 packs. Single-treated-unit uncertainty is large, though: synthetic DiD's placebo standard error is 11 packs. Code and data: https://github.com/yang-qs/stat5200-project.

1 Introduction

Between 1988 and 1992 two U.S. states passed ballot initiatives that combined a cigarette tax increase with an earmarked anti-smoking program. California's Proposition 99 (1988) raised the tax by 25 cents and funded media campaigns, school programs and local coalitions. Four years later Massachusetts' Question 1 did nearly the same thing: the tax rose from 26 to 51 cents per pack on 1 January 1993, and the revenue financed the Massachusetts Tobacco Control Program (MTCP). Whether such programs reduce smoking beyond what the tax alone would do is a recurring question in tobacco policy, and these two states are its standard test cases.

California's program is the canonical application of the synthetic control method. Abadie et al. (2010) build a "synthetic California" from 38 states without large tobacco programs and find that by 2000 per-capita sales were about 26 packs lower than they would have been without Proposition 99. Earlier evaluations of Massachusetts used simpler comparisons. Harris et al. (1996) compare Massachusetts with the other 48 contiguous states and report a faster decline in sales after 1992, and Biener et al. (2000) report similar trends using sales and survey data. Neither builds a counterfactual from pre-treatment data or quantifies uncertainty for a single treated state.

This project adds two things. First, it asks whether the California result generalises to the only other state that adopted a comparable program in that period, using the same data, donor pool and estimator, so that differences in the answer are not artefacts of method. Second, it puts the

synthetic control estimate next to two-way fixed effects DiD and the synthetic DiD estimator of Arkhangelsky et al. (2021). This shows how much the conclusion depends on the choice between DiD-type and SC-type estimators, and how wide honest single-unit confidence bands are. A by-product is an estimate of how much of the decline can plausibly be explained by the price increase alone.

2 Data

Outcome and prices. The main source is *The Tax Burden on Tobacco, 1970–2019* (Orzechowski and Walker, 2019), distributed by the CDC STATE System. It reports, for each state and fiscal year (ending 30 June), per-capita cigarette pack sales (computed from tax-paid sales), the average retail price per pack and the state and federal excise taxes. Abadie et al. (2010) use the same series. Because the data are by fiscal year, the January 1993 tax increase falls in the middle of fiscal 1993; following the convention Abadie et al. (2010) use for California's January 1989 increase, fiscal 1993 is the first treated year, and Section 4.4 shows that using 1994 instead changes nothing. The sample is 1970–2000: 23 pre-treatment and 8 post-treatment years for Massachusetts.

Covariates. To reproduce the predictors of Abadie et al. (2010) I merge three further state-year series: per capita personal income from the BEA (table SAINC1, logged); the share of the population aged 15–24, built from NCI SEER county population estimates; and beer consumption in gallons per resident aged 14 and over, from the NIAAA apparent-consumption file. Income is in nominal dollars. Since every state's predictors are averaged over the same years, a common price deflator cancels out of the comparison.

Donor pool. I use the donor pool of Abadie et al. (2010): all states except those that adopted large tobacco control programs (Arizona, California, Florida, Massachusetts and Oregon), those that raised cigarette taxes by 50 cents or more between 1989 and 2000 (Alaska, Hawaii, Maryland, Michigan, New Jersey, New York and Washington), and the District of Columbia. This leaves 38 donor states.

Table 1 and Figure 1 describe the data. Before 1993 Massachusetts sold about 11 packs per capita fewer than the average donor, and its sales were falling faster than the donor average during the late 1980s. That pre-trend is the main reason a simple comparison of means is not enough. The naive difference in mean changes is -19.3 packs.

Table 1: Means (standard deviations) over state-years.

	Massachusetts		Donor pool (38 states)	
	1970–92	1993–2000	1970–92	1993–2000
Packs per capita	115.56 (9.65)	70.20 (9.24)	126.23 (32.53)	100.20 (24.01)
Retail price ($/pack)	0.94 (0.46)	2.66 (0.71)	0.85 (0.44)	2.07 (0.54)
State tax ($/pack)	0.21 (0.04)	0.57 (0.18)	0.14 (0.07)	0.26 (0.14)
Tax share of price (%)	39.85 (9.43)	35.49 (4.67)	34.61 (9.67)	26.03 (6.54)

Tax share is federal plus state tax as a share of the retail price, recomputed from its components.

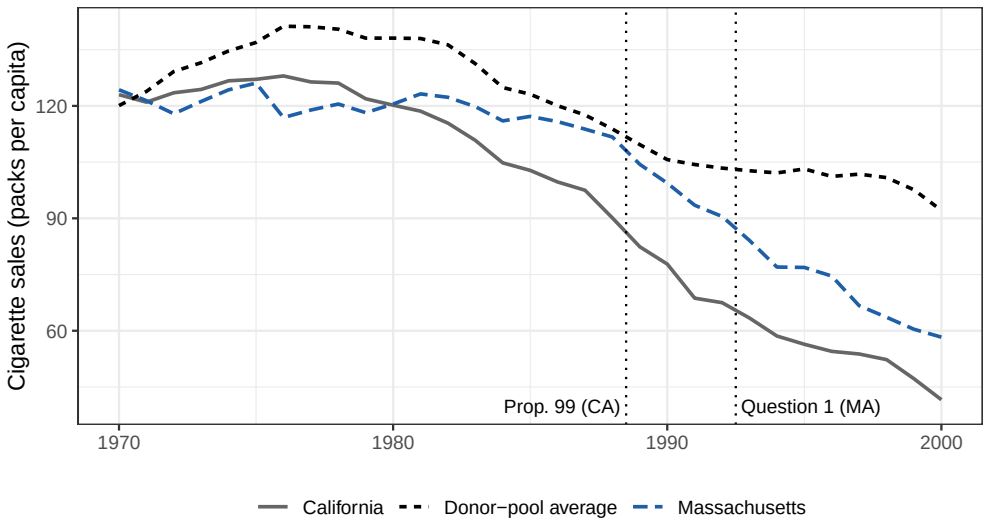


Figure 1: Per-capita cigarette sales, 1970–2000. Dotted lines mark the first treated years for California (1989) and Massachusetts (1993).

3 Models

Index states by $j = 1, \ldots, J+1$, with $j = 1$ the treated state and $j = 2, \ldots, J+1$ the $J = 38$ donors, and years by $t = 1, \ldots, T$, with T_0 the last untreated year. Let Y_{jt}^N be per-capita sales without the program and Y_{jt}^I sales with it. The parameters of interest are the yearly effects $\alpha_{1t} = Y_{1t}^I - Y_{1t}^N$ for $t > T_0$ and their average $\bar{\alpha} = (T - T_0)^{-1} \sum_{t>T_0} \alpha_{1t}$. Only Y_{1t}^I is observed after T_0, so each method is a way of imputing Y_{1t}^N.

3.1 Synthetic control

The synthetic control estimator (Abadie and Gardeazabal, 2003; Abadie et al., 2010) imputes $\hat{Y}_{1t}^N = \sum_{j \geq 2} w_j Y_{jt}$ with weights $w_j \geq 0$, $\sum_j w_j = 1$. Given a $k \times 1$ vector of pre-treatment predictors X_1 for the treated state and the matrix X_0 for donors, the weights solve

$$W^*(V) = \arg\min_{W} \, (X_1 - X_0 W)' V (X_1 - X_0 W), \tag{1}$$

and the diagonal matrix V is chosen to minimise the pre-treatment mean squared prediction error (MSPE) of the outcome. Following Abadie et al. (2010), the predictors are the averages of log income, retail price and the 15–24 population share over the nine years before treatment, average beer consumption over the five years before treatment, and per-capita sales in 1975, 1980 and the last pre-treatment year (1988 for California, 1992 for Massachusetts). The estimator is implemented with the `Synth` R package.

Inference uses placebos in space. Each donor in turn is treated as if it had adopted the program, with the other 38 states as its donor pool, and the ratio of post- to pre-treatment root MSPE is computed for each of the 39 units. The permutation p-value is the treated state's rank on this ratio divided by 39. Because the ratio uses squared gaps, it is two-sided and treats a large increase like a large decrease. I also report a one-sided version that ranks the signed statistic $\bar{\alpha}_j / \mathrm{RMSPE}_j^{\mathrm{pre}}$.

3.2 Two-way fixed effects DiD

The DiD benchmark is

$$Y_{jt} = \mu_j + \lambda_t + \tau\, D_{jt} + \varepsilon_{jt}, \qquad D_{jt} = \mathbf{1}\{j = 1,\ t > T_0\}, \tag{2}$$

estimated on the treated state and the 38 donors, which weights all donors equally and assumes parallel trends. With a single treated state, cluster-robust standard errors are not valid (Conley and Taber, 2011), so I report them only for reference and base inference on the same placebo-in-space permutation as for the synthetic control. I also estimate an event-study version with separate coefficients for each year relative to 1993, to see whether the parallel-trends assumption is plausible before 1993.

3.3 Synthetic difference-in-differences

Synthetic DiD (Arkhangelsky et al., 2021) combines the two:

$$(\hat\tau, \hat\mu, \hat\alpha, \hat\beta) = \arg\min \sum_j \sum_t \big(Y_{jt} - \mu - \alpha_j - \beta_t - \tau D_{jt}\big)^2 \hat\omega_j \hat\lambda_t, \tag{3}$$

where unit weights $\hat\omega$ make the weighted donor trend match the treated state's pre-treatment trend (not its level, because unit fixed effects remain) and time weights $\hat\lambda$ put weight on the pre-treatment years most similar to the post-treatment period. With $\hat\omega_j = 1/J$ and uniform $\hat\lambda$ the estimator is the DiD above. Without fixed effects it is a synthetic control using only outcomes. I estimate all three variants with the `synthdid` package and use its placebo variance estimator, the only one available with one treated unit.

3.4 Implied price response

Estimating the DiD model with $\log Y_{jt}$ and with the log retail price as outcomes gives the program's effect on log sales, τ_q, and on log price, τ_p. The ratio τ_q/τ_p is a Wald estimate of the price elasticity of demand, but only if the program affects sales through the price. Comparing it with elasticities from the literature indicates how much of the decline the tax alone can account for.

4 Results

All code is in the GitHub repository https://github.com/yang-qs/stat5200-project. `Rscript run_all.R` downloads the raw data and regenerates every number, table and figure in this report, and the README lists what each script does.

4.1 Replication: California

The replication reproduces the main results of Abadie et al. (2010). Synthetic California is a combination of Utah (0.37), Colorado (0.32), Nevada (0.24) and Connecticut (0.08); ADH report Utah, Nevada, Montana, Colorado and Connecticut. The predictors are well balanced (Table 2, left), and the pre-treatment RMSPE is 1.95 packs. The estimated gap in 2000 is -24.3 packs per capita, compared with about -26 in the paper, and California has the largest post/pre RMSPE

ratio of the 39 states, so the permutation p-value is $1/39 = 0.026$, as in the original. The small differences in weights and in the size of the gap are plausibly due to revisions of the tobacco series since 2010 and to my covariates being rebuilt from current releases of the BEA, SEER and NIAAA files. Figure 3 in the appendix shows the paths.

Table 2: Predictor balance: treated state, synthetic control and donor average.

	California	Synthetic	Donor average
Ln(per capita income)	9.65	9.49	9.42
Retail price ($/pack)	0.97	0.98	0.95
Share aged 15–24 (%)	17.32	17.04	17.30
Beer per capita (gal.)	29.79	30.03	30.27
Packs per capita, 1975	127.10	126.73	136.93
Packs per capita, 1980	120.20	120.44	138.09
Packs per capita, 1988	90.10	92.11	113.82

(a) California (1989)

	Massachusetts	Synthetic	Donor average
Ln(per capita income)	9.92	9.88	9.66
Retail price ($/pack)	1.44	1.44	1.32
Share aged 15–24 (%)	16.30	15.85	15.67
Beer per capita (gal.)	26.83	25.54	29.86
Packs per capita, 1975	126.10	119.03	136.93
Packs per capita, 1980	120.50	123.18	138.09
Packs per capita, 1992	90.50	90.98	103.41

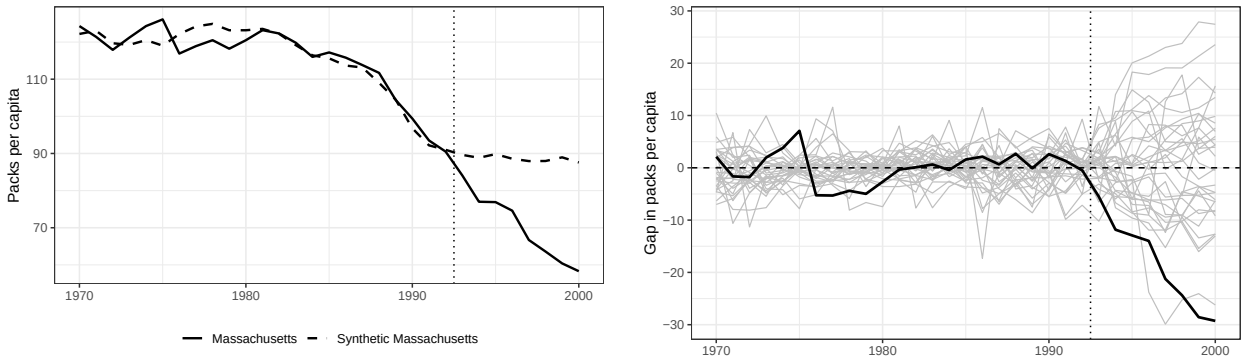
(b) Massachusetts (1993)

Covariates are averaged over the nine years before treatment, beer over the five years before treatment.

4.2 Synthetic Massachusetts

Synthetic Massachusetts puts weight 0.49 on Connecticut, 0.29 on Delaware, 0.15 on Utah and 0.07 on Rhode Island. It matches Massachusetts' income, price and 1992 sales closely and is far closer than the donor average on every predictor (Table 2, right). Its pre-treatment RMSPE is 3.0 packs, somewhat larger than California's, mostly because of a sharp one-year rise in Massachusetts' sales in 1975 that no combination of donors reproduces.

Figure 2(a) shows the result. From 1985 to 1992 the two series are almost indistinguishable. After 1992 synthetic Massachusetts levels off at about 88 packs, while actual sales keep falling, to 58.3 packs in 2000. The gap is -5.5 packs in 1993, -11.8 in 1994 and widens steadily to -29.3 by 2000. The average gap over 1993–2000 is -18.5 packs, 21% of counterfactual sales, which is almost the same proportional effect as California's (-17.8 packs, 23%).



(a) Massachusetts and synthetic Massachusetts

(b) Gaps: Massachusetts (black) and placebos (grey)

Figure 2: Synthetic control estimates for Massachusetts. Panel (b) omits placebos whose pre-treatment MSPE is more than five times that of Massachusetts.

Figure 2(b) compares the Massachusetts gap with the 38 placebo gaps. Massachusetts' gap is the most negative of any state by the late 1990s, but the two-sided test is less decisive than

for California. Missouri and Oklahoma have larger post/pre RMSPE ratios than Massachusetts, because their sales rose relative to their synthetic controls, so the two-sided permutation p-value is $3/39 = 0.077$. Ranking the signed standardised effect instead, no placebo shows a reduction as large as Massachusetts', giving a one-sided p-value of $1/39 = 0.026$. The one-sided test matches the hypothesis of interest (that the program reduced sales), but it was chosen after seeing that the two-sided test was weakened by placebo increases, so I treat 0.077 as the headline number.

4.3 Comparison with DiD and synthetic DiD

Table 3 reports average post-treatment effects from all estimators. For Massachusetts they lie between -15.7 and -19.3 packs, a narrow range. The agreement is less tight for California, where TWFE DiD (-27.4) is about 50% larger than the synthetic control estimate (-17.8): California's sales were already falling faster than the donor average before 1988, and equal donor weights attribute part of that pre-existing decline to Proposition 99. For Massachusetts the close agreement is partly a coincidence, as the event study below shows.

Table 3: Average effect on per-capita cigarette sales (packs) in the post-treatment period.

	California (1989)	Massachusetts (1993)
Two-way FE DiD	-27.4	-19.3
	(2.8)	(2.5)
	[0.128]	[0.179]
Synthetic control (ADH predictors)	-17.8	-18.5
	[0.026]	[0.077]
Synthetic control (outcomes only)	-19.6	-15.7
	(11.9)	(11.1)
Synthetic DiD	-15.2	-16.9
	(9.9)	(10.8)

Parentheses: cluster-robust (TWFE) or placebo (synthdid) standard errors. Brackets: placebo-in-space permutation p-values. TWFE p-values rank $|\hat{\tau}|$ across the 39 units; the synthetic control p-value ranks the post/pre RMSPE ratio. Rows 3 and 4 are from the `synthdid` package.

The table also shows how much the inferential method matters. TWFE cluster-robust standard errors of 2.5–2.8 packs would make every estimate look highly significant, but they are invalid with one treated cluster. The permutation p-values for TWFE are 0.13 (California) and 0.18 (Massachusetts): several donor states saw sales changes of similar size relative to the donor average. The synthdid placebo standard errors are 10–12 packs; the 95% interval for Massachusetts' synthetic DiD estimate is roughly $[-38, 4]$ packs. Synthetic control, which fits each state's own pre-period, is the only method under which Massachusetts stands out clearly from the placebos.

The event study (Figure 4 in the appendix) shows why parallel trends with equal donor weights is not credible for Massachusetts. Relative to 1992, the pre-treatment coefficients are far from zero and move in both directions: about $+17$ in 1970, between -12 and -5 in 1976–1980, and $+7$ to $+11$ in 1985–1990, falling to zero by 1992. Massachusetts was already losing sales faster than the average donor in the late 1980s, which on its own would make a DiD estimate too negative; the TWFE estimate averages over all of these pre-period deviations, and they partly offset each other. The post-1993 coefficients (-6 in 1993, about -24 by 1998–1999) are therefore hard to

read as effects. Synthetic control handles the problem by choosing donors whose late-1980s decline matches Massachusetts', which is why its estimate, rather than the DiD one, is the main result.

Price channel. Relative to the donors, Massachusetts' log retail price rose by 0.127 (about 13.5%) after 1992, and log sales fell by 0.272 (about 24%). The implied elasticity is -2.1 (Table 4); for California it is -3.4. Estimates of the price elasticity of cigarette demand cluster around -0.4 (Chaloupka and Warner, 2000). At that elasticity, a 13.5% price increase would reduce sales by roughly 5%, about a fifth of the observed decline. The rest has to come from channels other than price: the media campaign and other parts of the program, purchases moved to lower-tax states (which reduce recorded sales but not necessarily smoking), or changes in Massachusetts that the comparison does not capture.

Table 4: DiD effects on log sales and log price, and the implied elasticity.

	California	Massachusetts
DiD on log sales	-0.422	-0.272
DiD on log price	0.123	0.127
Implied price elasticity	-3.44	-2.14

4.4 Robustness

Table 5 reports the robustness checks.

In-time placebo. Pretending that the program started in 1988 and using only data up to 1992, the synthetic control (again mostly Connecticut and Delaware) gives an average gap of $+0.4$ packs over 1988–1992. The method does not create an effect where there should be none.

Neighbouring states. If Massachusetts smokers bought cigarettes in neighbouring states after the tax increase, those states' recorded sales would rise and inflate the counterfactual. Dropping Connecticut, New Hampshire, Rhode Island and Vermont from the donor pool makes the estimate larger, not smaller (-27.5 packs on average), so cross-border purchases recorded in the donor states do not explain the result. It cannot rule out purchases recorded in Massachusetts' other neighbour, New Hampshire, since New Hampshire has no weight in any specification.

Specification and timing. Using all 23 pre-treatment outcomes as predictors gives -15.8; making 1994, the first full fiscal year of the higher tax, the first treated year gives -18.4.

Leave-one-out. Dropping each donor with positive weight in turn gives average gaps between -13.7 (without Delaware or without Utah) and -26.9 (without Connecticut); Figure 5 in the appendix plots the paths. The sign never changes, but the size of the effect depends on the composition of the synthetic control by about a factor of two.

Table 5: Robustness of the Massachusetts synthetic control estimate (packs per capita).

Specification	Donors (weight ≥ 0.05)	Pre RMSPE	Avg. gap	Final gap
Baseline	CT (0.49), DE (0.29), UT (0.15), RI (0.07)	3.02	-18.5	-29.3
In-time placebo: 1988 (data to 1992)	CT (0.49), DE (0.24), WI (0.14), UT (0.09)	3.31	0.4	-1.7
Drop CT, NH, RI, VT from donors	DE (0.53), UT (0.31), MN (0.15)	3.75	-27.5	-41.0
All pre-period outcomes as predictors	CT (0.56), DE (0.13), OH (0.11), UT (0.09), RI (0.06)	2.97	-15.8	-24.2
First treated year 1994	CT (0.52), DE (0.24), UT (0.13), RI (0.11)	3.09	-18.4	-26.7
Drop CT	DE (0.50), UT (0.28), MN (0.22)	3.78	-26.9	-40.0
Drop DE	CT (0.67), IN (0.11), NM (0.09)	3.49	-13.7	-19.9
Drop RI	CT (0.51), DE (0.32), UT (0.15)	3.11	-19.8	-31.2
Drop UT	CT (0.67), IN (0.13), NM (0.10), RI (0.06)	3.45	-13.7	-20.0

Average gap over the post-treatment period; final gap in 2000 (in 1992 for the in-time placebo).

5 Conclusions and limitations

Massachusetts' 1992 tobacco control initiative was followed by a substantial fall in per-capita cigarette sales relative to a synthetic control that tracked the state closely for two decades. The average reduction over 1993–2000 is about 18 packs per capita (21%), and the gap widens to about 29 packs by 2000. In proportional terms this is almost the same effect that Abadie et al. (2010) estimate for California, which suggests that the California result was not a one-off. DiD and synthetic DiD point estimates agree, and the in-time placebo and leave-one-out checks support the sign of the effect.

The evidence is weaker than the point estimates make it look. The study has several limitations.

1. **One treated unit.** The two-sided permutation p-value is 0.077, synthetic DiD's 95% interval includes zero, and the leave-one-out estimates range from -14 to -27 packs. With a single state, the precision of the estimate is limited by the number of donors and by how well each one is fitted.

2. **The treatment is a bundle.** Question 1 raised the tax and funded the MTCP at the same time, and Massachusetts raised the tax again, to 76 cents, in 1996 (visible in the data from fiscal 1998). The estimate is the combined effect of all three. The price-channel calculation suggests that most of the decline came from something other than the 1993 price increase, but it cannot say whether that was the program, the later tax increase or tax avoidance.

3. **Sales are not smoking.** The outcome is tax-paid sales. Cross-border shopping, internet sales and smuggling reduce Massachusetts' recorded sales without reducing consumption; Lovenheim (2008) documents that this casual smuggling responds to tax differences between nearby states. The neighbour test rules out the case in which those purchases inflate the donors' sales, but not purchases from New Hampshire or from untaxed sources.

4. **Contaminated donors.** Several donors raised their own taxes after 1992 (Rhode Island from 37 to 71 cents, Connecticut from 40 to 50 cents, Utah by 25 cents in 1999), and many states began tobacco control activities after the 1998 Master Settlement Agreement. If anything, this lowers the counterfactual and biases the estimated effect toward zero in the last years of the sample.

5. **Measurement.** The O&W series is by fiscal year and the covariates by calendar year, and the covariates used here are current releases rather than the vintages used by Abadie et al. (2010). The replication shows that these differences matter little for California.

Natural extensions are individual-level outcomes (e.g. smoking prevalence from the CPS Tobacco Use Supplement or BRFSS) to separate smoking from sales, data on cross-border sales to quantify avoidance, and the augmented synthetic control estimator of Ben-Michael et al. (2021), which corrects for imperfect pre-treatment fit such as Massachusetts' 1975 spike.

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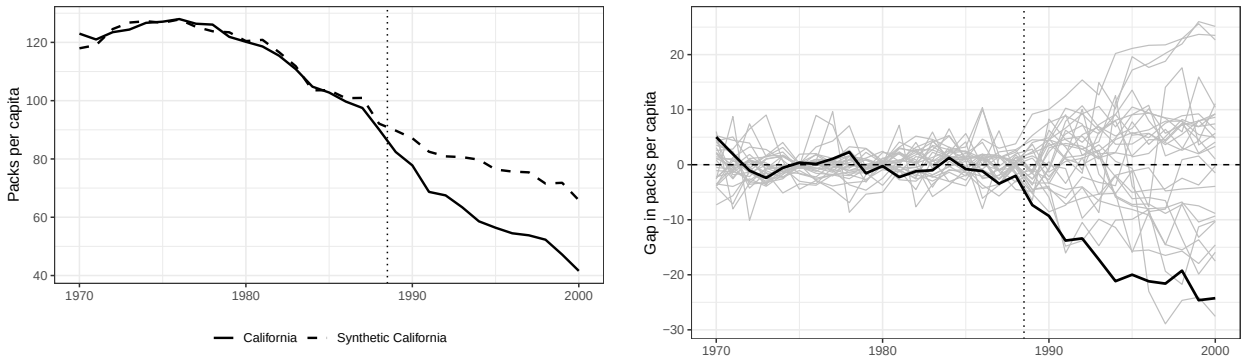
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A Additional figures



(a) California and synthetic California

(b) Gaps: California (black) and placebos (grey)

Figure 3: Replication of Abadie et al. (2010), Figures 2 and 5.

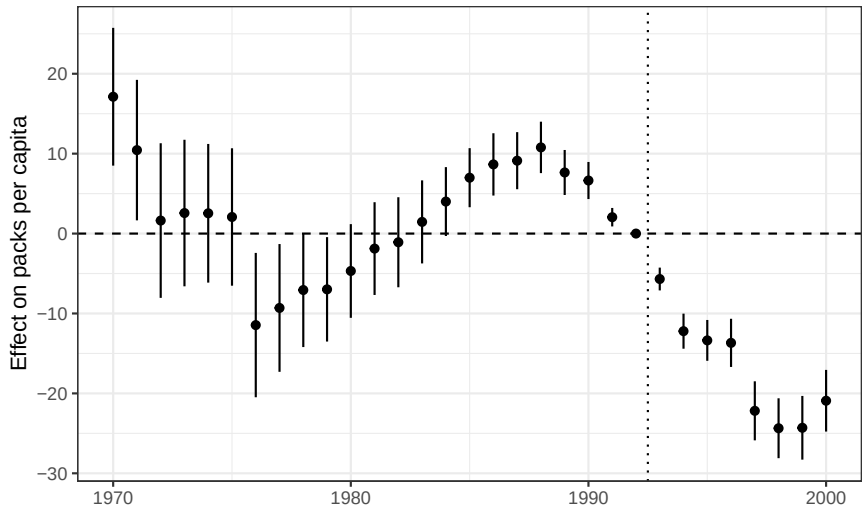


Figure 4: TWFE event study for Massachusetts, reference year 1992, with 95% cluster-robust intervals (which understate uncertainty with one treated state).

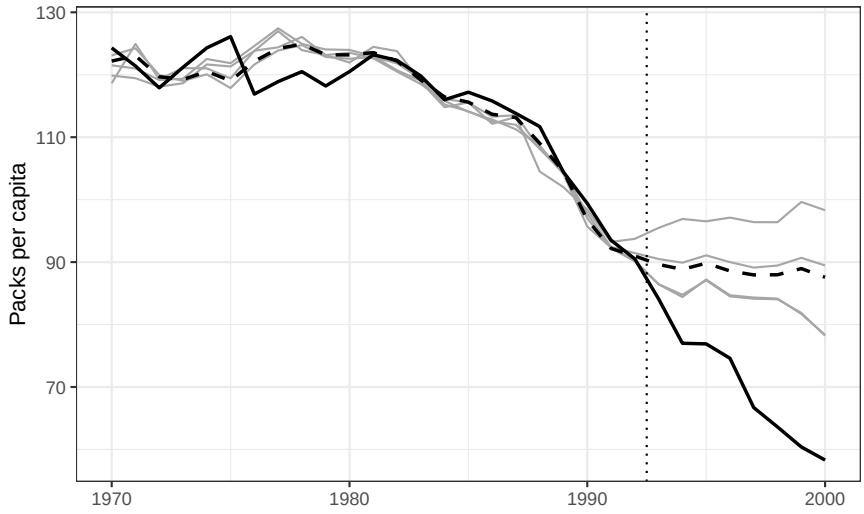


Figure 5: Leave-one-out: Massachusetts (solid), baseline synthetic control (dashed) and synthetic controls dropping one positive-weight donor at a time (grey).